Adam Patterson

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INTRODUCTION

Data Architect and Full-stack Engineer with **10+ years** of experience delivering solutions across diverse domains, including health sciences, education, mechanical systems, telephony, and **AI orchestration systems**. Holds a **Master's degree** in Applied Data Science from the University of Michigan.

EDUCATION

Master of Applied Data Science

University of Michigan – Ann Arbor (4.0/4.0)

Bachelor of Science

University of Michigan – Ann Arbor (3.6/4.0)

SKILLS

Programming Languages

C++, Java, **JavaScript**, **Node.js**, **Python,** PL/SQL, R, SQL, **TypeScript**

Cloud Computing

AWS ECS Fargate, **CloudFormation**, API Gateway, S3, AWS Lambda, IAM

Network Programming

HTTP, REST, RPC, and **WebSocket APIs**; TCP/IP; and secure networking protocols (SSL/TLS)

Geographic Information Systems

ArcGIS Pro (Python scripting)

Data Science & AI

Model fitting, evaluation, and imputation (Scikit-learn, PyTorch, etc.), visualization (ggplot2, Altair, Matplotlib, Plotly, etc.), data manipulation and big data processing (Pandas, Tidyverse, Spark, and MapReduce), telephony and VoIP, STT, TTS, and LLM model orchestrations (Telnyx, Twilio, Deepgram, Cartesia, 11Labs)

AWARDS

United States President's Volunteer Service Award

PUBLICATIONS

Coauthored 5 published articles in the health sciences domain.

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EMPLOYMENT

University of Michigan – Ann Arbor, MI Sept. 2017 to Present

Data Architect (current title)

- Advised senior principal investigators and senior administrators on secure data handling and AI adoption.
- Appointed as a departmental liaison with Information Assurance to ensure alignment with UM information security standards.
- Engineered a data integration system (Python/Pandas) using object-oriented design patterns that cleaned, transformed, and integrated institution wide data sources (SQL databases, HTTP APIs, and flat files) for the production of analysis datasets for scientific research.
- Applied data science (Pandas, Numpy, Scikit-learn, etc.), statistical methods, and advanced visualization methods (Plotly, Altair, D3) to further the research goals of the Laboratory.
- Ensured compliance with institutional HIPAA regulatory requirements.

Software Engineer (previous title)

- Developed full stack web applications (Python backend and JavaScript/TypeScript frontend), which were deployed to the Coursera open learning environment, to improve learning outcomes.
- Deployed scalable infrastructure using CloudFormation (e.g., JupyterHub using AWS ECS Fargate) to support research studies.
- Engineered full stack telemetry implementations that collected terabytes of data for scientific research (Python, TypeScript).
- Implemented cloud storage solutions that significantly reduced data transfer and storage costs for the Laboratory.
- Developed software that automated database migrations.
- Ensured compliance with FERPA regulations.

Washtenaw Community College Jan. 2015 to Sept. 2017

Systems Analyst

- Developed multiple full stack web applications (Java, JavaScript) including a key request system and an energy dashboard that integrated the college's 19 EIG Shark and Nexus Meters (Modbus API) for monitoring energy usage.
- Developed SQL queries and visualizations to monitor and improve work performance and inventory audits.
- Performed security audits on Division's Windows and Linux servers and devices and identified improvements to Division's IT security infrastructure.

PROJECTS

GitHub: github.com/adamjpatterson

Developed several open-source projects, including:

- **Dialog:** VoIP STT, TTS, and LLM orchestration framework
- **Socketnaut:** network server scaling framework
- **Streams Logger:** type-safe logging pipeline built on native Node.js streams
- **Network-Services:** type-safe bi-directional RPC facility
- **Persistence:** filesystem coordination layer
- **Scalability:** type-safe service scaling facility using lightweight worker threads